Experiment No. 3

```
# Undo and Redo System using Stack

undo_stack = [ ]
redo_stack = [ ]

document = " "

def make_change(new_text):
        global document

        # Store current state before making a change
        undo_stack.append(document)

        # Make the new change
        document = new_text

        # Clear redo stack because a new change is made
        redo_stack.clear()

        print("Change made successfully.")

def undo():
        global document

        if len(undo_stack) == 0:
                print("Nothing to undo.")
        else:
                # Store current state in redo stack
                redo_stack.append(document)

                # Get previous state from undo stack
                document = undo_stack.pop()

                print("Undo performed successfully.")

def redo():
        global document

        if len(redo_stack) ==0:
                print("Nothing to redo.")
        else:
                # Store current state in undo stack
                undo_stack.append(document)

                # Get the most recently undone state
                document = redo_stack.pop()

                print("Redo performed successfully.")

def display_document():
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    print("\nCurrent Document State:")
    print(document)

# Main program

while True:

    print("\n-------TEXT EDITOR-------")
    print("1. Make a Change")
    print("2. Undo")
    print("3. Redo")
    print("4. Display Document")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        new_text = input("Enter new document text: ")
        make_change(new_text)

    elif choice == "2":
        undo()

    elif choice == "3":
        redo()

    elif choice == "4":
        display_document()

    elif choice == "5":
        print("Program terminated.")
        break

    else:
        print("Invalid choice.")
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